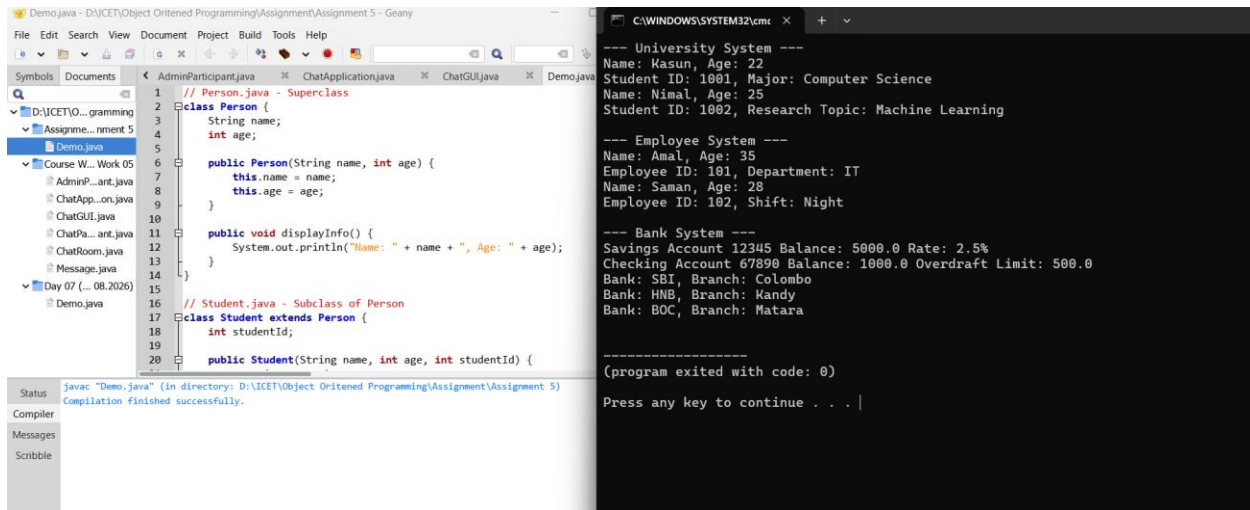


ASSIGNMENT 05

PR26121077

01)

- **Classes:** Employee, Student, Person, Account, Bank
- **Super**
Classes: Employee (for Manager, Worker), Student (for GraduateStudent, Undergraduate Student), Person (for Student, Employee), Account (for SavingsAccount, CheckingAccount), Bank (for SBI, HNB, BOC)
- **Sub**
Classes: Manager, Worker, GraduateStudent, UndergraduateStudent, SavingsAccount, CheckingAccount, SBI, HNB, BOC



The screenshot shows a Java IDE with two windows. The left window displays the source code for `Person.java` and `Student.java`. `Person.java` is the superclass with attributes `String name` and `int age`, and a `displayInfo()` method. `Student.java` is a subclass that extends `Person` and adds an `int studentId` attribute. The right window is a terminal showing the output of the program, which displays information for a University System, an Employee System, and a Bank System.

```
// Person.java - Superclass
class Person {
    String name;
    int age;

    public Person(String name, int age) {
        this.name = name;
        this.age = age;
    }

    public void displayInfo() {
        System.out.println("Name: " + name + ", Age: " + age);
    }
}

// Student.java - Subclass of Person
class Student extends Person {
    int studentId;

    public Student(String name, int age, int studentId) {
        super(name, age);
    }
}
```

```
--- University System ---
Name: Kasun, Age: 22
Student ID: 1001, Major: Computer Science
Name: Nimal, Age: 25
Student ID: 1002, Research Topic: Machine Learning

--- Employee System ---
Name: Amal, Age: 35
Employee ID: 101, Department: IT
Name: Saman, Age: 28
Employee ID: 102, Shift: Night

--- Bank System ---
Savings Account 12345 Balance: 5000.0 Rate: 2.5%
Checking Account 67890 Balance: 1000.0 Overdraft Limit: 500.0
Bank: SBI, Branch: Colombo
Bank: HNB, Branch: Kandy
Bank: BOC, Branch: Matara

(program exited with code: 0)
Press any key to continue . . . |
```

02)

Inheritance is a mechanism where a new class (subclass/child) derives properties and behaviors from an existing class (superclass/parent). It represents an "is-a" relationship.

Real-World Examples:

- **Vehicle Hierarchy:** Vehicle is the superclass. Car, Bike, and Truck are subclasses. All vehicles have wheels and an engine, but cars have four doors, bikes have two wheels, etc.

- **Employee Hierarchy:** Employee is the superclass. Manager, Developer, and Intern are subclasses. All employees have a name and salary, but managers have a team to manage, developers write code, etc.
- **Animal Hierarchy:** Animal is the superclass. Dog, Cat, and Bird are subclasses. All animals eat and sleep, but dogs bark, cats meow, and birds fly.

03)

How Inheritance Promotes Software Reuse, Saves Time, and Prevents Errors:

1. **Promotes Software Reuse:** Common code (attributes and methods) is written once in the superclass. This code is automatically reused by all subclasses without rewriting it. This eliminates redundancy.
2. **Saves Time During Development:** Developers don't need to start from scratch. They can quickly create new classes by extending existing, well-tested ones. This reduces the overall development time.
3. **Helps Prevent Errors:**
 - **Reduced Code Duplication:** Bugs often arise from duplicate code. If a bug is found in the superclass, it is fixed in one place, and all subclasses automatically inherit the fix.
 - **Reuse of Tested Code:** By reusing existing code that has already been tested and debugged, the risk of introducing new bugs is significantly reduced.

04)

// Case 1: Student Hierarchy

```
class Student { /* ... */ }
class GraduateStudent extends Student { /* ... */ }
class UndergraduateStudent extends Student { /* ... */ }
```

// Case 2: Shape Hierarchy

```
class Shape { /* ... */ }
class Circle extends Shape { /* ... */ }
class Triangle extends Shape { /* ... */ }
```

```
class Rectangle extends Shape { /* ... */ }
```

```
class Sphere extends Shape { /* ... */ }
```

```
class Cube extends Shape { /* ... */ }
```

```
// Case 3: Loan Hierarchy
```

```
class Loan { /* ... */ }
```

```
class CarLoan extends Loan { /* ... */ }
```

```
class HomeImprovementLoan extends Loan { /* ... */ }
```

```
class MortgageLoan extends Loan { /* ... */ }
```

```
// Case 4: Employee Hierarchy
```

```
class Employee { /* ... */ }
```

```
class Faculty extends Employee { /* ... */ }
```

```
class Staff extends Employee { /* ... */ }
```

```
// Case 5: BankAccount Hierarchy
```

```
class BankAccount { /* ... */ }
```

```
class CheckingAccount extends BankAccount { /* ... */ }
```

```
class SavingsAccount extends BankAccount { /* ... */ }
```

05)

```
// Expand the Shape hierarchy:
```

```
class Shape {
```

```
    // Generic Shape attributes like color, position
```

```
    public void draw() { System.out.println("Drawing a shape."); }
```

```
    public double area() { return 0; }
```

```
}
```

```
// Two-Dimensional Shapes
```

```
class TwoDimensionalShape extends Shape {  
    // Common 2D attributes like area calculation method  
    @Override  
    public double area() { return 0; } // to be overridden by specific 2D shapes  
}
```

```
class Circle extends TwoDimensionalShape {  
    private double radius;  
    public Circle(double radius) { this.radius = radius; }  
    @Override  
    public double area() { return Math.PI * radius * radius; }  
}
```

```
class Square extends TwoDimensionalShape {  
    private double side;  
    public Square(double side) { this.side = side; }  
    @Override  
    public double area() { return side * side; }  
}
```

```
class Triangle extends TwoDimensionalShape { /* ... */ }
```

```
class Rectangle extends TwoDimensionalShape { /* ... */ }
```

```
// Three-Dimensional Shapes
```

```
class ThreeDimensionalShape extends Shape {
```

```
// Common 3D attributes like volume calculation method
public double volume() { return 0; }
}

class Sphere extends ThreeDimensionalShape {
    private double radius;

    public Sphere(double radius) { this.radius = radius; }

    @Override
    public double area() { return 4 * Math.PI * radius * radius; } // Surface area

    @Override
    public double volume() { return (4.0/3.0) * Math.PI * radius * radius * radius; }
}

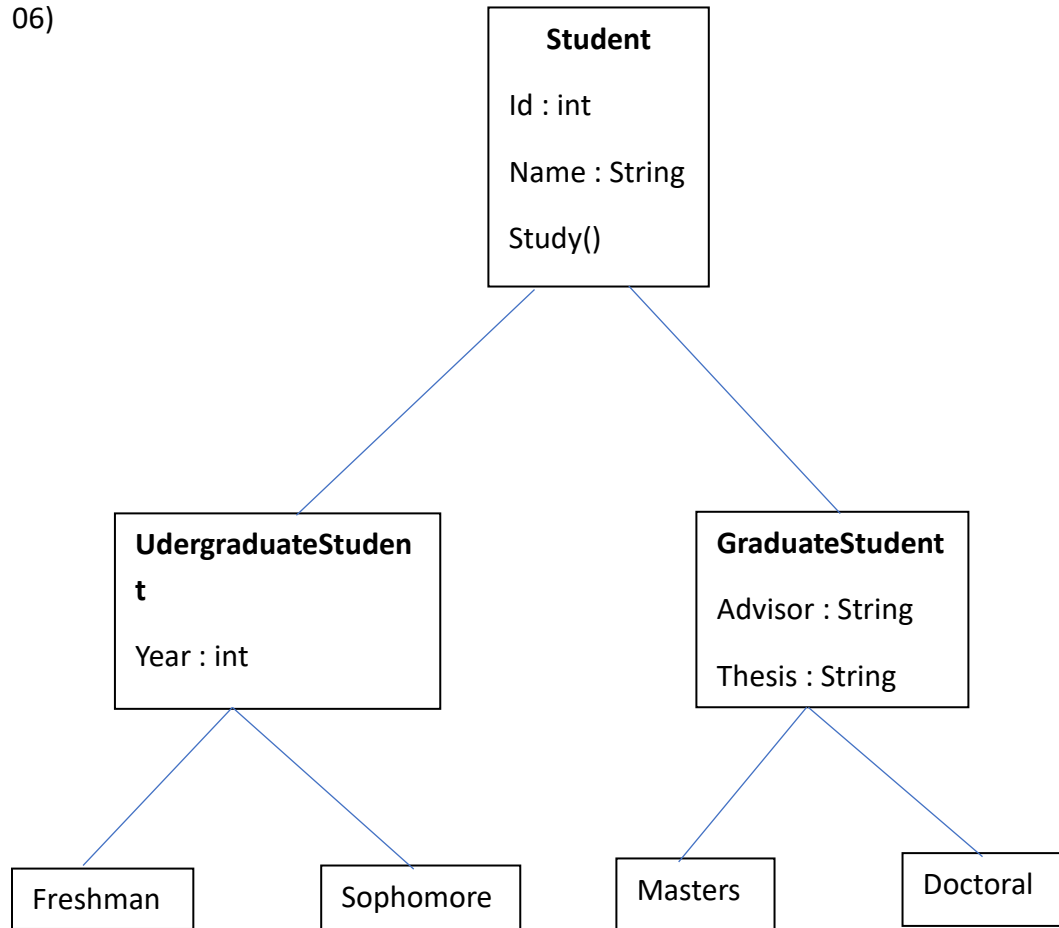
class Cube extends ThreeDimensionalShape {
    private double side;

    public Cube(double side) { this.side = side; }

    @Override
    public double area() { return 6 * side * side; } // Surface area

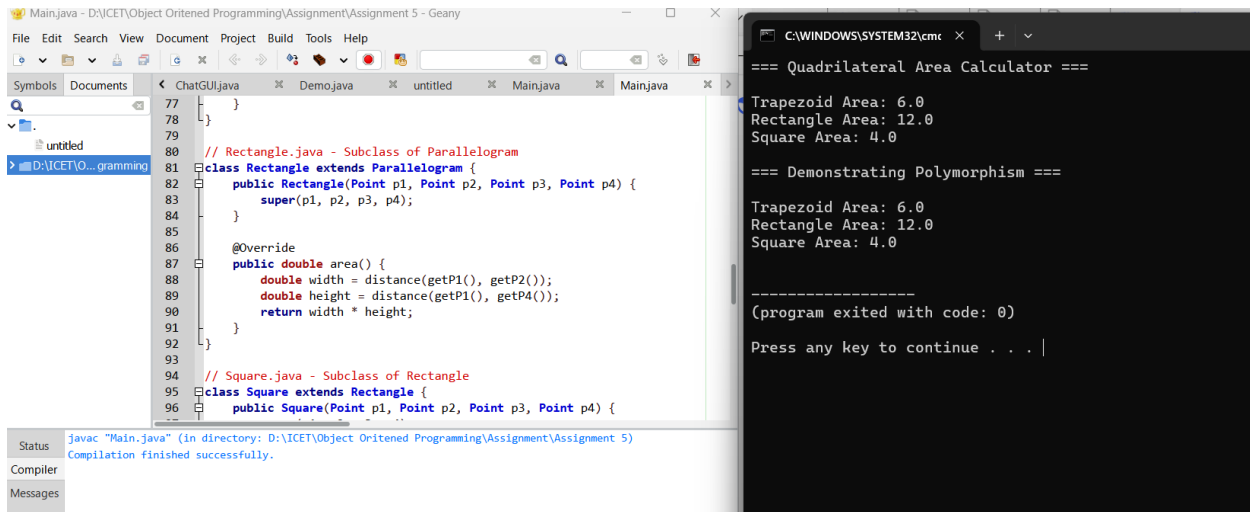
    @Override
    public double volume() { return side * side * side; }
}
```

06)



```
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File Edit Search View Document Project Build Tools Help
Symbols Documents
untitled
> D:\JCE\O...gramming
82 // MastersStudent.java - Subclass of GraduateStudent
83
84 class MastersStudent extends GraduateStudent {
85     public MastersStudent(int id, String name, String advisor, String thesis) {
86         super(id, name, advisor, thesis);
87     }
88 }
89
90 // DoctoralStudent.java - Subclass of GraduateStudent
91
92 class DoctoralStudent extends GraduateStudent {
93     public DoctoralStudent(int id, String name, String advisor, String thesis) {
94         super(id, name, advisor, thesis);
95     }
96 }
97
98 // Main.java - Public class with matching filename
99
100 public class Main {
101     public static void main(String[] args) {
102         System.out.println("=== University Student System ===\n");
103     }
104 }
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07)



The screenshot shows an IDE window titled 'Main.java - D:\ICET\Object Oriented Programming\Assignment\Assignment 5 - Geany'. The code defines a `Rectangle` class extending `Parallelogram` and a `Square` class extending `Rectangle`. The `Rectangle` class has a constructor taking four `Point` objects and an `area()` method. The `Square` class has a constructor taking four `Point` objects. The output window shows the results of running the program: 'Trapezoid Area: 6.0', 'Rectangle Area: 12.0', 'Square Area: 4.0', and 'Demonstrating Polymorphism'.

```
// Rectangle.java - Subclass of Parallelogram
class Rectangle extends Parallelogram {
    public Rectangle(Point p1, Point p2, Point p3, Point p4) {
        super(p1, p2, p3, p4);
    }

    @Override
    public double area() {
        double width = distance(getP1(), getP2());
        double height = distance(getP1(), getP4());
        return width * height;
    }
}

// Square.java - Subclass of Rectangle
class Square extends Rectangle {
    public Square(Point p1, Point p2, Point p3, Point p4) {
        super(p1, p2, p3, p4);
    }
}
```

```
=== Quadrilateral Area Calculator ===

Trapezoid Area: 6.0
Rectangle Area: 12.0
Square Area: 4.0

=== Demonstrating Polymorphism ===

Trapezoid Area: 6.0
Rectangle Area: 12.0
Square Area: 4.0

-----
(program exited with code: 0)
Press any key to continue . . .
```

08)

Software Reusability in OOP using the Java Application:

- CalculatorView is a subclass of JFrame. By extending JFrame, it **reuses** all the pre-built functionality of a window in Java Swing (like setTitle, setSize, setVisible, setDefaultCloseOperation).
- This is a classic example of **Inheritance**. Without writing any complex code to create a window from scratch, CalculatorView automatically inherits all the methods and properties of JFrame.
- This saves development time, reduces errors, and promotes code reuse. The Demo class then creates an instance of CalculatorView and uses the inherited methods to configure and display the window.

09)

True Statements: A, B, C, D, E, F

- A: True. Inheritance promotes reuse.
- B: True. Java supports multiple inheritance of type (interfaces) but not of implementation.
- C: True. That's the definition of single inheritance.

- D: True. A subclass specializes the superclass.
- E: True. This is the foundation of polymorphism.
- F: True. A 'has-a' relationship is composition.

10)

Compile Error: Lines 4 and 5 in B.java.

- **Line 4:** printA(); will cause a compile error because class B does not extend class A, so B has no knowledge of the printA() method.
- **Line 5:** System.out.println("a:"+a); will cause a compile error because class B does not have a variable a. a is a member of class A.

11)

True Statements: B and C

- B: True. The relationship between a subclass and its superclass is an "is-a" relationship (e.g., Dog **is a** Animal).
- C: True. A class containing a reference to another class is a "has-a" relationship (e.g., Car **has a** Engine).

12)

True Statements: A, B, C, D

- A: True. Private members are not inherited.
- B: True. Constructors are not inherited.
- C: True. super(...) must be the first statement in a subclass constructor.
- D: True. This is method overriding.
- E: False. When a subclass redefines a method with the same signature (name and parameters), it is called **overriding**, not overloading. Overloading is when the parameters are different.

13)

Compile Statements: A, B

- A: Correct. This is method overriding with the exact same signature.

- B: Correct. The parameter name is different, but the signature (method name + parameter types) is the same. Java only cares about the parameter types, not the parameter names.
- C: Incorrect. Parameter type is double, so this is method overloading, not overriding. It will compile.
- D: Incorrect. Parameter type is int[], so this is method overloading, not overriding. It will compile.

Note: The question asks "Which statement(s), inserted at line 12, will compile". All four will compile. However, if the question specifically intended to refer to "overriding" (same signature), then only A and B correctly override the method.

14)

Compilation Error.

Explanation: Class B extends class A. Class A has a **parameterized constructor** A(int i). When a class doesn't have a default constructor (no-arg), the compiler does not provide one.

Class B implicitly tries to call a default constructor of A (super()), which doesn't exist, causing a compile-time error.

15)

Compile Error Lines: 9, 10, 12, 13, 14, 15, 16, 17

Explanation:

- **Inheritance Hierarchy:** A (superclass) B extends A C extends A D extends B.
- **Reference Variable Rule:** A superclass reference variable can hold an object of any subclass (upcasting). A subclass reference variable **cannot** hold a superclass object.
- **Line 9:** B b2 = a1; - B (subclass) trying to reference a1 (A object). **Error.**
- **Line 10:** b2 = c1; - B and C are siblings (both extend A). No relation. **Error.**
- **Line 11:** b2 = d1; - D is a subclass of B, so B b2 = d1; is allowed.
- **Line 12:** C c2 = a1; - C (subclass) trying to reference a1 (A object). **Error.**
- **Line 13:** c2 = b1; - C and B are siblings. **Error.**
- **Line 14:** c2 = d1; - D is not a subclass of C. **Error.**

- **Line 15:** D d2 = a1; - **Error.**
- **Line 16:** d2 = b1; - **Error.**
- **Line 17:** d2 = c1; - **Error.**

16)

Correct Answer: B

Explanation: Overriding is when a subclass provides a specific implementation for a method that is already defined in its superclass. The subclass method has the **same name, same parameters, and same return type** (or covariant). It "replaces" the superclass method's functionality for the subclass objects.

17)

Compile Statements: D

- A: Incorrect. If Animal doesn't extend Shape it's a circular dependency or undefined.
- B: Incorrect. This creates a circular inheritance (Shape extends Animal while Animal is outside).
- C: Incorrect. Java does not support multiple inheritance of classes.
- D: Correct. This creates a new class AnimalShape.

18)

Correct Answers: B, C, E

Explanation: Dynamic method dispatch (runtime polymorphism) is the mechanism where a call to an overridden method is resolved at runtime based on the **type of the object** the reference variable points to, not the type of the reference variable.

- A: a1.myMethod(100); - a1 refers to A object. Calls A's method.
- B: a2.myMethod(100); - a2 is A reference, but points to B object. Calls B's method. **(Runtime Polymorphism)**
- C: a3.myMethod(100); - a3 is A reference, points to C object. Calls C's method. **(Runtime Polymorphism)**
- D: b1.myMethod(100); - b1 refers to B object. Calls B's method.

- E: b2.myMethod(100); - b2 is B reference, points to C object. Calls C's method. (**Runtime Polymorphism**)
- F: c1.myMethod(100); - c1 refers to C object. Calls C's method.

19)

Compilation Error.

Explanation: Class Super does not have a default constructor. Class Sub extends Super, and the Sub constructor `public Sub(int i){ System.out.println("Sub(int)"); }` implicitly tries to call the default constructor of Super (`super()`), which does not exist. The Sub class must explicitly call the parameterized constructor of Super using `super(i);`.

20)

Compile Statements: E, F

Explanation: The `super(...)` call must be the first statement in the constructor. The arguments must be valid and accessible from the context of the constructor.

- A: `super(a);` - a is an instance variable of Super. It cannot be used in a `super()` call because `super()` must be the first statement, and instance variables of the superclass are not accessible to the subclass before the superclass constructor is called.
- B: `super(b);` - b is an instance variable of Sub. It cannot be used before the superclass constructor is called.
- C: `super(x);` - x is a static variable of Super. It can be used. The code will compile.
- D: `super(y);` - y is not defined anywhere.
- E: `super(i);` - i is the constructor parameter. It can be used. The code will compile.
- F: `super(100);` - This directly passes a value. The code will compile.

21)

Mechanism of Invoking a Superclass Constructor:

A subclass constructor must call its superclass's constructor to ensure the superclass part of the object is properly initialized. This is done using the `super` keyword.

- **Rule:** The `super(...)` call must be the **first statement** in the subclass constructor's body.

- **Implicit Call:** If the subclass constructor does not explicitly call `super(...)`, the compiler inserts an implicit call to the superclass's **default (no-argument) constructor** (`super()`). If the superclass doesn't have a default constructor, a compile-time error occurs.
- **Explicit Call:** The subclass must explicitly call `super(parameters)` to invoke the parameterized constructor of its superclass.

22)

```

class Parent {

    int value = 10;

}

class Child extends Parent {

    int value = 20; // This hides Parent's value

    void display() {

        System.out.println(value);    // 20

        System.out.println(super.value); // 10

    }

}

```

23) **F. It is not possible to invoke test() method defined in C from a method in A.**

24) **Answer: F. hi hi, followed by an exception**

25)

26)

Figure figref;

```
figref = r; // Rectangle object
```

```
System.out.println("Area is " + figref.area()); // Calls Rectangle's area()
```

```
figref = t; // Triangle object
```

```
System.out.println("Area is " + figref.area()); // Calls Triangle's area()
```

```
figref = f; // Figure object
```

```
System.out.println("Area is " + figref.area()); // Calls Figure's area()
```

27) A. ABDABCE

28)

29) **D. none of the above**

30) **A, B, C are TRUE. D is FALSE.**

31)

33) **B, E, F will cause errors.**

34)

```
import java.util.*;
```

```
class Customer {
```

```
    private int code;
```

```
    private String name;
```

```

public Customer(int code, String name) {
    this.code = code;
    this.name = name;
}

public int getCode() { return code; }
public String getName() { return name; }

@Override
public String toString() {
    return code + "- " + name;
}
}

class CustomerStack {
    private List<Customer> stack = new ArrayList<>();

    public void push(Customer c) {
        stack.add(c);
    }

    public Customer pop() {
        if (!stack.isEmpty()) {
            return stack.remove(stack.size() - 1);
        }
        return null;
    }
}

```

```
}
```

```
public void printCustomerStack() {  
    List<String> result = new ArrayList<>();  
    for (int i = stack.size() - 1; i >= 0; i--) {  
        result.add(stack.get(i).toString());  
    }  
    System.out.println(result);  
}
```

```
}
```

35) **Answer: A. ABC**

36) **B and C are TRUE.**

37) **D. Prints Sub : null → Panadura**